



Wrocław
University
of Science
and Technology

Software System Development (Level 2) (Lecture)

Overview of managerial activities

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Agenda

Classification of the functions of management

Planning

- Types of business plans

- Planning methods

- Methods and integrated planning systems

- Financial planning

Organizing

- Basic concepts of organizing

- Basic methods of organizing

Coordination

Human Resources Management

- Personnel administration

- Wages and Salaries

- Personnel

 - Levels of hierarchy

 - Formal Organizational Structure



1

Consultations

Consultations

Consultations: Summer 2026

- Tuesday:

1:00 p.m. – **3:00** p.m. (13:00 – 15:00)

Building: **D2**, Room: **302b**

- Friday:

1:00 p.m. – **3:00** p.m. (13:00 – 15:00)

Zoom platform

The consultations during my office hours are by appointment only. The e-mail with the date and hour of choice should be send at least **2** days in advance with the specific topic of the consultations.

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2

Final test

Final test

Exam: Summer 2026

- I term:

Date: **29 June** 2026

Time: **3:15** p.m. – 3:40 p.m. (15:15 – 15:40)

- II term:

Date: **6 July** 2026

Time: **3:15** p.m. – 3:40 p.m. (15:15 – 15:40)

Rules:

- The exam will take place online.
- It will be possible to join the exam only within the first 5 minutes of its start time.
- To take the exam, you need access to the Internet, a computer and an installed web browser.
- No additional dates are planned.
- The exam result is expressed as a percentage and constitutes an element of the assessment for a group of courses, in accordance with the rules presented below.



3

Subject objectives



Subject objectives

- To familiarize students with managerial activities.



4

Classification of the functions of management

Classification of the functions of management

Classification according to *Henri Fayol*:

- Planning
- Organizing
- Commanding
- Controlling
- Coordinating

Classification of the functions of management

Classification according to *Luther Gulick* and *Lyndall Urwick*:

- Planning
- Organizing
- Staffing
- Directing
- Coordinating
- Reporting
- Budgeting

Classification of the functions of management

Classification according to *Heinz Weihrich* and *Harold Koontz*:

- Planning
- Organizing
- Human Resources Management
- Leadership
- Control

Classification of the functions of management

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5

Planning

Planning

"The supreme task of business management is to identify specific business goals through which an ultimate goal, long-term profit maximization is to be achieved, as well as the formulation of business policy, i.e. 'a marching route' which the enterprise has to adhere to in order to achieve these goals as efficiently as possible."

Günther Wöhe

Planning

- **Strategic** planning,
- **Tactical** planning,
- **Operational** planning.



5.1

Planning

Types of business plans



Planning

Types of business plans

Types of business plans:

- Project plan
- Production plan
- Human resources development plan
- Sales plan
- Personal Development Plan
- Investment plan
- Maintenance and repair plan
- Financial plan (integrates all plans with the help of a financial statement)



5.2

Planning

Planning methods

Planning

Planning methods

Planning methods:

- Bidirectional Planning (Top-Down/Bottom-Up)
- Progressive Planning (Bottom-Up)
- Retrograde Planning (Top-Down)
- Strategic Alternatives
- Strategic Management



5.3

Planning

Methods and integrated planning systems



Planning

Methods and integrated planning systems

Methods and integrated planning systems:

- APS (Advanced Planning and Scheduling)
- ERP (Enterprise Resource Planning)
- MRP (Material Requirements Planning)
- MRP II (Manufacturing Resource Planning)
- JIT (Just-in-time)
- KANBAN
- DBR (Drum Buffer Rope)
- SCM (Supply Chain Management)



5.4

Planning

Financial planning



Planning

Financial planning

Financial planning:

- Financial result planning

$$\textit{Profit} = \textit{Revenues} - \textit{Costs}$$



Planning

Financial planning

Financial planning:

- Revenue planning



Planning

Financial planning

Financial planning:

- Costs planning



Planning

Financial planning

Financial planning:

- Capital planning



Planning

Financial planning

Financial planning:

- Cash flow planning

Term *cash flow* can be seen as:

- free money supply that is available to the business (static concept),
- future detachable revenue that an investor can get when investing to the business (dynamic concept).



6

Organizing

Organizing

The reasons for organizing:

- division of labor,
- delegation of authority and responsibility,
- setting span of control.



6.1

Organizing

Basic concepts of organizing



Organizing

Basic concepts of organizing

Basic concepts of organizing:

- Adhocracy – a flexible, adaptable, and informal organizational structure without bureaucratic policies or procedures.



Organizing

Basic concepts of organizing

Basic concepts of organizing:

- Bureaucracy – a system of government in which most of the important decisions are made by state officials rather than by elected representatives.



Organizing

Basic concepts of organizing

Basic concepts of organizing:

- Meritocracy – government or the holding of power by people selected on the basis of their ability.



6.2

Organizing

Basic methods of organizing

Organizing

Basic methods of organizing

Basic methods of organizing:

- Centralization – management and decision-making powers are concentrated

Organizing

Basic methods of organizing

Basic methods of organizing:

- Decentralization – management and decision making powers are delegated

Organizing

Basic methods of organizing

Basic methods of organizing:

- Empowerment – focuses on strengthening the powers of workers or organizational units

Organizing

Basic methods of organizing

Basic methods of organizing:

- Organizational Development – increasing the effectiveness and viability of the organization



Organizing

Basic methods of organizing

Basic methods of organizing:

- Working group – social group that occurs in the workplace

Organizing

Basic methods of organizing

Basic methods of organizing:

- OSCAR – acronym for a method of organizing designed by *Ernest Dale*.

Organizing process must include:

- Objectives
- Specialization
- Coordination
- Authority
- Responsibility

Organizing

Basic methods of organizing

Basic methods of organizing:

OSCAR method

- **O**bjectives – it is necessary to establish and mutually logically link the objectives of the organizational units, objectives (outputs) of the processes and objectives of the work on individual jobs



Organizing

Basic methods of organizing

Basic methods of organizing:

OSCAR method

- **Specialization** – processes, jobs and organizational units should be grouped based on the similarity of the activities carried out



Organizing

Basic methods of organizing

Basic methods of organizing:

OSCAR method

- **Coordination** – activities and processes must be mutually linked and aligned to lead from the time and place aspect to the most logical and efficient interconnection



Organizing

Basic methods of organizing

Basic methods of organizing:

OSCAR method

- **Authority** – the proper functioning of the activities and processes requires correct setting of authority so that any activity, process, position and department should have adequately allocated power

Organizing

Basic methods of organizing

Basic methods of organizing:

OSCAR method

- **R**esponsibility – correct functioning of activities and processes requires correct setting of responsibilities so that at each organizational unit and position is clear what is responsible



7

Coordination



Coordination

Coordination is one of the key managerial functions and thus applies to all aspects of the organization: Economy and Finance, Computer Science, Quality, Human Resources, Logistics and Transport, Organizational Management, Marketing, Services and Production.



Coordination

The essence of **coordination** is to ensure the material, financial, temporal and possibly spatial **coherence** across the enterprise's objectives, its parts, activities or individual workers.



8

Human Resources Management

Human Resources Management

Human Resources (HR) Management is an area of processes in an organization that deals with a human resources management and development.

It thus encompasses all management methods and analytical techniques applied in human resources management and in the work with people in the organization – from staff recruitment, employment contract closing to wage payment.

Human Resources Management

Human Resources Management includes:

- personnel administration,
- wages and salaries,
- personnel.



8.1

Human Resources Management

Personnel administration



Human Resources Management

Personnel administration

Personnel administration:

- Personal Data
- Personal File
- State Obligations
- Information for organization needs
- Staff requirements
- Employment contract/Labour relations



8.2

Human Resources Management

Wages and Salaries

Human Resources Management

Wages and Salaries

Wages and Salaries:

- Wage records and systematization
- Worktime Fund
- Employee Remuneration
- Incentive components of wages
- Payroll Accounting
- Payroll Processing
- Record of Hours Worked
- Wages Settlement and Payment



8.3

Human Resources Management

Personnel



Human Resources Management

Personnel

Personnel:

- Levels of hierarchy
- Management parts
- Organizational structure
- Organizational Chart
- Organigram



Human Resources Management

Personnel

Levels of hierarchy

Levels of hierarchy – Levels in the management hierarchy is a term that characterizes the number of degrees of organizational units within the organizational structure of the organization.

Organizational structure:

- Tall organizational structure – high number of management levels
- Flat organizational structure – low number of management levels

Human Resources Management

Personnel

Formal Organizational Structure

Formal Organizational Structure – officially codified hierarchical arrangement of relationships between different jobs within the organizational units and relationships between departments within the organization (shown in a *Organigram*).

Human Resources Management

Personnel

Formal Organizational Structure

The formal organizational structure of an organization is a type of a social network. Depending on the type and size, there are different typologies of formal organizational structures.

The formal organizational structure includes:

- Functional Organizational Structure
- Linear Organizational Structure
- Matrix Organizational Structure
- Staff and Line Organizational Structure



Human Resources Management

Personnel

Formal Organizational Structure

Functional organizational structure – an arrangement where a worker has different managers for different areas of the organization operation.

Human Resources Management

Personnel

Formal Organizational Structure

Linear organizational structure – the position and relations of superiority and inferiority are arranged and oriented vertically.



Human Resources Management

Personnel

Formal Organizational Structure

Matrix organizational structure – (also *project structure*) a classic vertical linear structure, which is combined with a horizontal structure showing ad-hoc generated teams dedicated to special projects.

The matrix organizational structure is necessary in project-oriented organizations.

Human Resources Management

Personnel

Formal Organizational Structure

Staff and Line organizational structure – based on the arrangement of linear organizational structure extended by the staff departments that provide support for management activities for different hierarchical levels and areas of the operation of the organization.

Thank you for your attention